Minhyek Jeon (minhyek.jeon@cs.ox.ac.uk)
University of Oxford, Department of Computer Science.
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EDUCATION

University of Oxford (Advisor: João F. Henriques, Irina Voiculescu) *Oct.2025 – Current*
Doctor of Philosophy in Computer Science

Carnegie Mellon University (Advisor: Fernando De la Torre) *Aug.2023 – May.2025*
Master of Science in Computational Biology
Graduated with Honors in Academic Achievement and Research Excellence
Thesis Title: Towards Generalizable Chest X-Ray Diagnosis via Stochastic Interpolants
• GPA of 4.03/4.33

Korea University *Mar.2017 – Aug.2023*
Bachelor of Engineering in Artificial Intelligence, Bachelor of Science in Biotechnology
Dean's Fellowship: 2022 Summer
• GPA of 4.04/4.5

PUBLICATION

Conference Proceedings:

[p.7] **M Jeon**, M Conn ,S Abram, I Voiculescu, Self-Supervised Radiograph Canonicalization via 3D Geometry-Guided De-rotation for Knee Osteotomy, Submitted to ***WACV*****,** 2026.

[p.8] **S** Panev[1], **M Jeon**[1], C de Melo, S Hu, S Chakraborty, F De la Torre, Diagnosing Aerial-View Object Detectors with Foundational Image Generative Models, accepted to ***ECCV*****,** 2026, (Co-first author).

[p.6] X Fang, **M Jeon**, Z Qin, S Panev, C de Melo, S Hu, S Chakraborty, F De la Torre, Adapting vehicle detectors for aerial imagery to unseen domains with weak supervision, ***ICCV***, 2025.

[p.5] R Aggarwal, D Peñaherrera, J Shao, **M Jeon**, D Koes, Models trained with unnormalized density functions: A need for a course correction, ***ICLR Blogposts***, 2025.

Journal Articles:

[p.4] HG Jeong[1], **M Jeon[1]**, KJ Ryu, J Kim, BY Choe, YY Joo, H Park, Similar but distinct comorbidity patterns between polycystic ovary syndrome and endometriosis in Korean women: A nationwide cohort study, ***Journal of Korean Medical Science***, 2024, (Co-first author).

[p.3] S Lee, Y Cho, Y Ji, **M Jeon**, A Kim, YY Joo, Multimodal integration of neuroimaging and genetic data for the diagnosis of mood disorders based on computer vision models, ***Journal of Psychiatric Research***, 2024.

[p.2] J Kubica, R Kumar, G Tan, VQ Truong, D Enoma, N Cooley, **M Jeon**, et al. Biomedical data management, knowledge graphs, and deep learning, ***OSF,*** 2023.

[p.1] Y Kim[1], Y Lim[1], **M Jeon[1]**, S Kim, K Kwon, S Hwang, J Lee, Modeling of metabolic syndrome outbreaks due to climate crisis and reconsidering the direction of international development cooperation in Eritrea, ***Journal of Africa Future Society***, 2023, (Co-first author).

RESEARCH EXPERIENCE

University of Oxford: Oxford, United Kingdom
Oct.2025 to Present

[p.8] VGG Group (DPhil Researcher): Developed generative modeling frameworks for medical imaging to produce anatomically realistic data for model training. Investigated bias detection, reliability, and interpretability to improve diagnostic model generalization in clinical applications.

Carnegie Mellon University: Pittsburgh, Pennsylvania, USA
Jan.2024 to Oct.2025

[p.6,7] Human Sensing Lab (Research Assistant): Collaborated with U.S. Department of Defense on cross-domain generalization and model interpretability for vision tasks. Fine-tuned text-to-image diffusion models for synthetic data generation, achieving 12.8% AP50 improvement in object detection. Developed counterfactual frameworks to diagnose model biases across visual domains.

Sep.2023 to Dec.2023

Robert Murphy Lab (Research Assistant): Developed automated image analysis pipeline for segmentation and tracking of biomolecular fusion events in time-lapse microscopy data, with a focus on monitoring the dynamics and merging behavior of ALT-associated promyelocytic leukemia nuclear bodies.

University of Pittsburgh: Pittsburgh, Pennsylvania, USA
Jan.2024 to Dec.2024

[p.5] David Ryan Koes Lab (Research Assistant): Evaluated iterative energy matching frameworks against Sequential Monte Carlo methods for equilibrium sampling from Boltzmann distributions. Demonstrated that traditional MCMC methods achieve superior convergence efficiency and ensemble property prediction compared to state-of-the-art neural samplers across high-dimensional

molecular systems (LJ-13, LJ-55).

Korea University: Seoul, Republic of Korea
Mar.2022 to Aug.2023

[p.4] Institute of Data Science (Undergraduate Research Assistant): Worked with Department of Obstetrics and Gynecology at Korea University Anam Hospital to analyze 157,662 Korean women clinical data. Conducted phenome-wide association studies using multivariate logistic regression to identify comorbidity patterns between polycystic ovary syndrome and endometriosis, demonstrating diametric pathophysiological profiles supporting inverse comorbidity mechanisms.

Mar.2021 to Mar.2022

Computer Vision Lab (Undergraduate Research Assistant): Developed confidence-guided semi-supervised framework for semantic correspondence leveraging tri-modal trainable confidence estimation networks. Addressed intra-class variations and background clutter through transformer-based global reasoning, outperforming baseline methods by 6.3%.

WORK EXPERIENCE

Babyscape: Pittsburgh, Pennsylvania, USA
Dec.2024 to Jan.2026

Co-founder & CTO: Founded an AI-driven diagnostic platform leveraging deep learning algorithms for automated fetal ultrasound image analysis to determine fetal sex, assess developmental milestones, and identify congenital anomalies. Successfully accepted into the Carnegie Mellon Swartz Center accelerator program under the mentorship of Craig Markovitz.

Korea University Anam Hospital, Department of Radiology: Seoul, Republic of Korea
Sep.2022 to Aug.2023

[p.3] Research Associate: Developed first multimodal fusion architecture integrating neuroimaging and genomic modalities for mood disorder diagnosis. Leveraged T1-weighted structural MRI sequences and whole-exome sequencing with polygenic risk scores, achieving superior diagnostic performance demonstrating 32.03% improvement over SNP-only and 8.19% improvement over image-only baseline models.

World Federation of United Nations Association: New York, USA
Apr.2023 to Aug.2023

[p.1] Korean Student Delegation to the UN Headquarters: Conducted quantitative analysis of climate-health correlations in Eritrea using WHO data on 131 disease types. Applied linear regression and covariance analysis to identify strong correlation between temperature rise/deforestation and diabetes prevalence.

Korea National Police Agency: Seoul, Republic of Korea
Apr.2018 to Dec.2019

English-Korean Interpreter: Fulfilled mandatory military service as an English-Korean interpreter for official residences of ambassadors from Turkey, Japan, Saudi Arabia, and China, and assisted with international affairs at the Seongbuk Police Station.

TEACHING EXPERIENCE

- **Internal Supervisor,** Group Design Practical for Computer Science, Mathematics Undergraduates *Jan.2026 - Jun.2026*
- **Practical Demonstrator,** Artificial Intelligence, University of Oxford *Jan.2026 - Mar.2026*
- **Class Tutor,** Computer Vision, University of Oxford *Oct.2025 - Dec.2025*
- **Summer Teaching Assistant,** AI Scholars (AI4All), Carnegie Mellon University *Jun.2025 - July.2025*
- **Teaching Assistant,** Machine Learning for Scientists, Carnegie Mellon University *Jan.2025 - May.2025*
- **Teaching Assistant,** Data Structure, Korea University *Mar.2022 - Jun.2022*
- **Instructor,** Samsung Dream Class, Samsung, Taught and mentored high school students in STEM *Mar.2019 - Feb.2020*

AWARDS&HONORS

- **Graduate Development Scholarship, St Anne's College, University of Oxford (£7,864)** *June.10th, 2026*
- **Overseas Student Scholarship, Mogam Science Scholarship Foundation ($10,000)** *Nov.28th, 2025*
- **2026 Fulbright Scholarship for Advanced STEM Fields, KAEC ($90,000)** *Sep.19th, 2025*
  Selected as the principal candidate for the Fulbright Scholarship; declined as it was tenable only at U.S. universities.
- **Certificate of Excellence in Research, Carnegie Mellon University** *May.10th, 2025*
  Awarded for distinctive research performance in the M.S. program in Computational Biology.
- **Certificate of Excellence in Academic Achievement, Carnegie Mellon University** *May.10th, 2025*
  Awarded in recognition of graduating at the top of the class in the M.S. program in Computational Biology.
- **Nucleate BioHack, Patient Safety Technology Challenge Award** *Oct.25th, 2024*
  As Project leader, led a project on medical image domain adaptation, selected as one of the top 10 projects.
- **Semester High Honors at Division of Biotechnology, Korea University** *2020, 2022*
  Received awards (Spring 2020, Spring & Fall 2022) for ranking among the top students in academic performance.
- **Future Leaders Scholarship, Korea University ($2,650)** *Sep.1st, 2022*
  Awarded for distinguished performance as a student representative at the United Nations, in recognition of global engagement.
- **Award Certificate, Seoul Metropolitan Police Agency** *Oct. 21st, 2019*
- **KAIST Presidential Award, 12th Chemistry Frontier Festival** *Oct.31st, 2015*